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Calculus on Manifolds

A MODERN APPROACH TO CLASSICAL THEOREMS
OF ADVANCED CALCULUS

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Functions on Euclidean Space

NORM AND INNER PRODUCT

Euclidean n-space $\mathbf{R}^n$ is defined as the set of all n-tuples $(x^1, \ldots, x^n)$ of real numbers x^i (a "1-tuple of numbers" is just a number and $\mathbf{R}^1 = \mathbf{R}$, the set of all real numbers). An element of $\mathbf{R}^n$ is often called a point in $\mathbf{R}^n$, and $\mathbf{R}^1$, $\mathbf{R}^2$, $\mathbf{R}^3$ are often called the line, the plane, and space, respectively. If x denotes an element of $\mathbf{R}^n$, then x is an n-tuple of numbers, the ith one of which is denoted x^i; thus we can write

$$x = (x^1, \ldots, x^n).$$

A point in $\mathbf{R}^n$ is frequently also called a vector in $\mathbf{R}^n$, because $\mathbf{R}^n$, with $x + y = (x^1 + y^1, \ldots, x^n + y^n)$ and $ax = (ax^1, \ldots, ax^n)$, as operations, *is* a vector space (over the real numbers, of dimension n). In this vector space there is the notion of the length of a vector x, usually called the **norm** $|x|$ of x and defined by $|x| = \sqrt{(x^1)^2 + \cdots + (x^n)^2}$. If $n = 1$, then $|x|$ is the usual absolute value of x. The relation between the norm and the vector space structure of $\mathbf{R}^n$ is very important.

1-1 Theorem. *If $x,y \in \mathbf{R}^n$ and $a \in \mathbf{R}$, then*

(1) $|x| \geq 0$, *and* $|x| = 0$ *if and only if* $x = 0$.

(2) $|\Sigma_{i=1}^n x^i y^i| \leq |x| \cdot |y|$; *equality holds if and only if x and y are linearly dependent.*

(3) $|x + y| \leq |x| + |y|$.

(4) $|ax| = |a| \cdot |x|$.

Proof

(1) is left to the reader.

(2) If x and y are linearly dependent, equality clearly holds. If not, then $\lambda y - x \neq 0$ for all $\lambda \in \mathbf{R}$, so

$$\begin{aligned} 0 < |\lambda y - x|^2 &= \sum_{i=1}^n (\lambda y^i - x^i)^2 \\ &= \lambda^2 \sum_{i=1}^n (y^i)^2 - 2\lambda \sum_{i=1}^n x^i y^i + \sum_{i=1}^n (x^i)^2. \end{aligned}$$

Therefore the right side is a quadratic equation in λ with no real solution, and its discriminant must be negative. Thus

$$4\left(\sum_{i=1}^n x^i y^i\right)^2 - 4\sum_{i=1}^n (x^i)^2 \cdot \sum_{i=1}^n (y^i)^2 < 0.$$

(3) $$\begin{aligned} |x + y|^2 &= \Sigma_{i=1}^n (x^i + y^i)^2 \\ &= \Sigma_{i=1}^n (x^i)^2 + \Sigma_{i=1}^n (y^i)^2 + 2\Sigma_{i=1}^n x^i y^i \\ &\leq |x|^2 + |y|^2 + 2|x| \cdot |y| \qquad \text{by (2)} \\ &= (|x| + |y|)^2. \end{aligned}$$

(4) $|ax| = \sqrt{\Sigma_{i=1}^n (ax^i)^2} = \sqrt{a^2 \Sigma_{i=1}^n (x^i)^2} = |a| \cdot |x|$. ▮

The quantity $\Sigma_{i=1}^n x^i y^i$ which appears in (2) is called the **inner product** of x and y and denoted $\langle x,y \rangle$. The most important properties of the inner product are the following.

1-2 Theorem. *If x, x_1, x_2 and y, y_1, y_2 are vectors in $\mathbf{R}^n$ and $a \in \mathbf{R}$, then*

(1) $\langle x,y \rangle = \langle y,x \rangle$ *(symmetry).*

(2) $\langle ax,y\rangle = \langle x,ay\rangle = a\langle x,y\rangle$ (*bilinearity*).
$\langle x_1 + x_2, y\rangle = \langle x_1,y\rangle + \langle x_2,y\rangle$
$\langle x, y_1 + y_2\rangle = \langle x,y_1\rangle + \langle x,y_2\rangle$

(3) $\langle x,x\rangle \geq 0$, *and* $\langle x,x\rangle = 0$ *if and only if* $x = 0$ (*positive definiteness*).

(4) $|x| = \sqrt{\langle x,x\rangle}$.

(5) $\langle x,y\rangle = \dfrac{|x + y|^2 - |x - y|^2}{4}$ (*polarization identity*).

Proof

(1) $\langle x,y\rangle = \Sigma_{i=1}^n x^iy^i = \Sigma_{i=1}^n y^ix^i = \langle y,x\rangle$.

(2) By (1) it suffices to prove

$$\langle ax,y\rangle = a\langle x,y\rangle,$$
$$\langle x_1 + x_2, y\rangle = \langle x_1,y\rangle + \langle x_2,y\rangle.$$

These follow from the equations

$$\langle ax,y\rangle = \sum_{i=1}^n (ax^i)y^i = a\sum_{i=1}^n x^iy^i = a\langle x,y\rangle,$$
$$\begin{aligned}\langle x_1 + x_2, y\rangle &= \sum_{i=1}^n ({x_1}^i + {x_2}^i)y^i = \sum_{i=1}^n {x_1}^iy^i + \sum_{i=1}^n {x_2}^iy^i \\ &= \langle x_1,y\rangle + \langle x_2,y\rangle.\end{aligned}$$

(3) and (4) are left to the reader.

(5) $$\begin{aligned}&\frac{|x + y|^2 - |x - y|^2}{4} \\ &= \tfrac{1}{4}[\langle x + y, x + y\rangle - \langle x - y, x - y\rangle] \qquad \text{by (4)} \\ &= \tfrac{1}{4}[\langle x,x\rangle + 2\langle x,y\rangle + \langle y,y\rangle - (\langle x,x\rangle - 2\langle x,y\rangle + \langle y,y\rangle)] \\ &= \langle x,y\rangle. \quad \blacksquare\end{aligned}$$

We conclude this section with some important remarks about notation. The vector $(0, \ldots ,0)$ will usually be denoted simply 0. The **usual basis** of $\mathbf{R}^n$ is $e_1, \ldots ,e_n$, where $e_i = (0, \ldots ,1, \ldots ,0)$, with the 1 in the ith place. If $T: \mathbf{R}^n \to \mathbf{R}^m$ is a linear transformation, the matrix of T with respect to the usual bases of $\mathbf{R}^n$ and $\mathbf{R}^m$ is the $m \times n$ matrix $A = (a_{ij})$, where $T(e_i) = \Sigma_{j=1}^m a_{ji}e_j$ —the coefficients of $T(e_i)$

appear in the ith *column* of the matrix. If $S\colon \mathbf{R}^m \to \mathbf{R}^p$ has the $p \times m$ matrix B, then $S \circ T$ has the $p \times n$ matrix BA [here $S \circ T(x) = S(T(x))$; most books on linear algebra denote $S \circ T$ simply ST]. To find $T(x)$ one computes the $m \times 1$ matrix

$$\begin{pmatrix} y^1 \\ \cdot \\ \cdot \\ \cdot \\ y^m \end{pmatrix} = \begin{pmatrix} a_{11}, & \ldots & ,a_{1n} \\ \cdot & & \cdot \\ \cdot & & \cdot \\ \cdot & & \cdot \\ a_{m1}, & \ldots & ,a_{mn} \end{pmatrix} \cdot \begin{pmatrix} x^1 \\ \cdot \\ \cdot \\ \cdot \\ x^n \end{pmatrix};$$

then $T(x) = (y^1, \ldots, y^m)$. One notational convention greatly simplifies many formulas: if $x \in \mathbf{R}^n$ and $y \in \mathbf{R}^m$, then (x,y) denotes

$$(x^1, \ldots, x^n, y^1, \ldots, y^m) \in \mathbf{R}^{n+m}.$$

Problems. **1-1.*** Prove that $|x| \le \sum_{i=1}^n |x^i|$.

1-2. When does equality hold in Theorem 1-1(3)? *Hint:* Re-examine the proof; the answer is not "when x and y are linearly dependent."

1-3. Prove that $|x - y| \le |x| + |y|$. When does equality hold?

1-4. Prove that $\big|\, |x| - |y| \,\big| \le |x - y|$.

1-5. The quantity $|y - x|$ is called the **distance** between x and y. Prove and interpret geometrically the "triangle inequality": $|z - x| \le |z - y| + |y - x|$.

1-6. Let f and g be integrable on $[a,b]$.

(a) Prove that $\left|\int_a^b f \cdot g\right| \le (\int_a^b f^2)^{\frac{1}{2}} \cdot (\int_a^b g^2)^{\frac{1}{2}}$. *Hint:* Consider separately the cases $0 = \int_a^b (f - \lambda g)^2$ for some $\lambda \in \mathbf{R}$ and $0 < \int_a^b (f - \lambda g)^2$ for all $\lambda \in \mathbf{R}$.

(b) If equality holds, must $f = \lambda g$ for some $\lambda \in \mathbf{R}$? What if f and g are continuous?

(c) Show that Theorem 1-1(2) is a special case of (a).

1-7. A linear transformation $T\colon \mathbf{R}^n \to \mathbf{R}^n$ is **norm preserving** if $|T(x)| = |x|$, and **inner product preserving** if $\langle Tx,Ty \rangle = \langle x,y \rangle$.

(a) Prove that T is norm preserving if and only if T is inner-product preserving.

(b) Prove that such a linear transformation T is 1-1 and T^{-1} is of the same sort.

1-8. If $x,y \in \mathbf{R}^n$ are non-zero, the **angle** between x and y, denoted $\angle(x,y)$, is defined as $\arccos(\langle x,y \rangle / |x| \cdot |y|)$, which makes sense by Theorem 1-1(2). The linear transformation T is **angle preserving** if T is 1-1, and for $x,y \ne 0$ we have $\angle(Tx,Ty) = \angle(x,y)$.

(a) Prove that if T is norm preserving, then T is angle preserving.

(b) If there is a basis $x_1, \ldots, x_n$ of $\mathbf{R}^n$ and numbers $\lambda_1, \ldots, \lambda_n$ such that $Tx_i = \lambda_i x_i$, prove that T is angle preserving if and only if all $|\lambda_i|$ are equal.

(c) What are all angle preserving $T\colon \mathbf{R}^n \to \mathbf{R}^n$?

1-9. If $0 \leq \theta < \pi$, let $T\colon \mathbf{R}^2 \to \mathbf{R}^2$ have the matrix $\begin{pmatrix} \cos\theta, & \sin\theta \\ -\sin\theta, & \cos\theta \end{pmatrix}$. Show that T is angle preserving and if $x \neq 0$, then $\angle(x,Tx) = \theta$.

1-10.* If $T\colon \mathbf{R}^m \to \mathbf{R}^n$ is a linear transformation, show that there is a number M such that $|T(h)| \leq M|h|$ for $h \in \mathbf{R}^m$. *Hint:* Estimate $|T(h)|$ in terms of $|h|$ and the entries in the matrix of T.

1-11. If $x,y \in \mathbf{R}^n$ and $z,w \in \mathbf{R}^m$, show that $\langle (x,z),(y,w)\rangle = \langle x,y\rangle + \langle z,w\rangle$ and $|(x,z)| = \sqrt{|x|^2 + |z|^2}$. Note that (x,z) and (y,w) denote points in $\mathbf{R}^{n+m}$.

1-12.* Let $(\mathbf{R}^n)^*$ denote the dual space of the vector space $\mathbf{R}^n$. If $x \in \mathbf{R}^n$, define $\varphi_x \in (\mathbf{R}^n)^*$ by $\varphi_x(y) = \langle x,y\rangle$. Define $T\colon \mathbf{R}^n \to (\mathbf{R}^n)^*$ by $T(x) = \varphi_x$. Show that T is a 1-1 linear transformation and conclude that every $\varphi \in (\mathbf{R}^n)^*$ is φ_x for a unique $x \in \mathbf{R}^n$.

1-13.* If $x,y \in \mathbf{R}^n$, then x and y are called **perpendicular** (or **orthogonal**) if $\langle x,y\rangle = 0$. If x and y are perpendicular, prove that $|x+y|^2 = |x|^2 + |y|^2$.

SUBSETS OF EUCLIDEAN SPACE

The closed interval $[a,b]$ has a natural analogue in $\mathbf{R}^2$. This is the **closed rectangle** $[a,b] \times [c,d]$, defined as the collection of all pairs (x,y) with $x \in [a,b]$ and $y \in [c,d]$. More generally, if $A \subset \mathbf{R}^m$ and $B \subset \mathbf{R}^n$, then $A \times B \subset \mathbf{R}^{m+n}$ is defined as the set of all $(x,y) \in \mathbf{R}^{m+n}$ with $x \in A$ and $y \in B$. In particular, $\mathbf{R}^{m+n} = \mathbf{R}^m \times \mathbf{R}^n$. If $A \subset \mathbf{R}^m$, $B \subset \mathbf{R}^n$, and $C \subset \mathbf{R}^p$, then $(A \times B) \times C = A \times (B \times C)$, and both of these are denoted simply $A \times B \times C$; this convention is extended to the product of any number of sets. The set $[a_1,b_1] \times \cdots \times [a_n,b_n] \subset \mathbf{R}^n$ is called **a closed rectangle** in $\mathbf{R}^n$, while the set $(a_1,b_1) \times \cdots \times (a_n,b_n) \subset \mathbf{R}^n$ is called an **open rectangle.** More generally a set $U \subset \mathbf{R}^n$ is called **open** (Figure 1-1) if for each $x \in U$ there is an open rectangle A such that $x \in A \subset U$.

A subset C of $\mathbf{R}^n$ is **closed** if $\mathbf{R}^n - C$ is open. For example, if C contains only finitely many points, then C is closed.

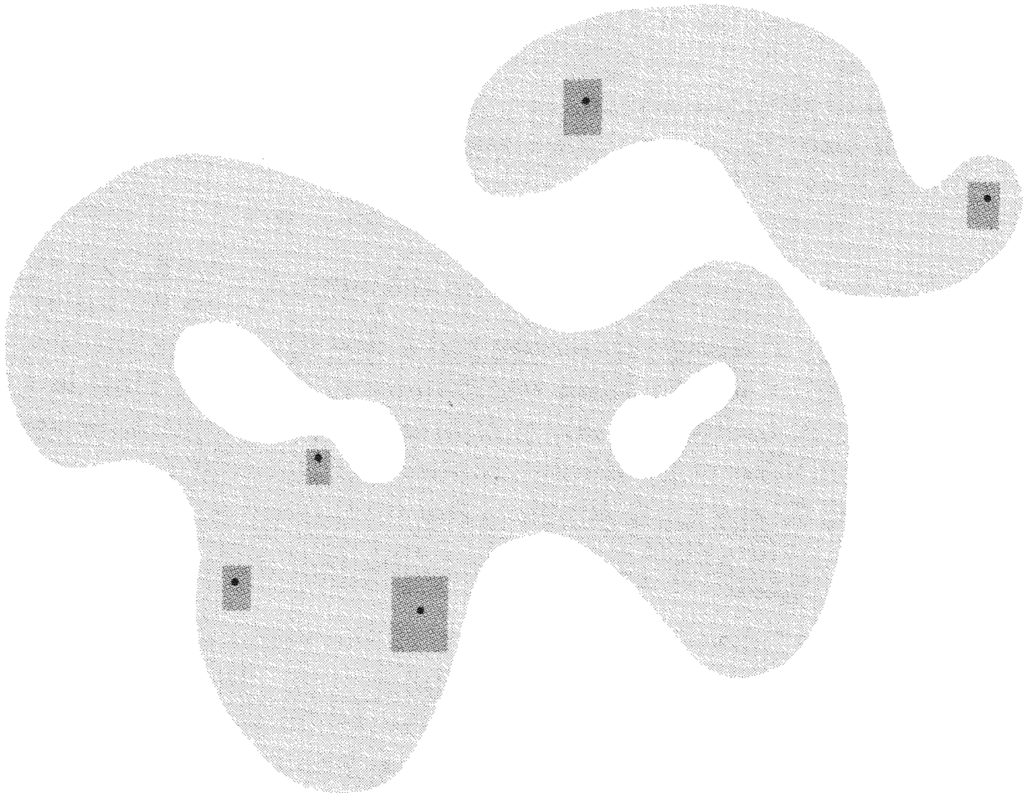

FIGURE 1-1

The reader should supply the proof that a closed rectangle in $\mathbf{R}^n$ is indeed a closed set.

If $A \subset \mathbf{R}^n$ and $x \in \mathbf{R}^n$, then one of three possibilities must hold (Figure 1-2):

1. There is an open rectangle B such that $x \in B \subset A$.
2. There is an open rectangle B such that $x \in B \subset \mathbf{R}^n - A$.
3. If B is any open rectangle with $x \in B$, then B contains points of both A and $\mathbf{R}^n - A$.

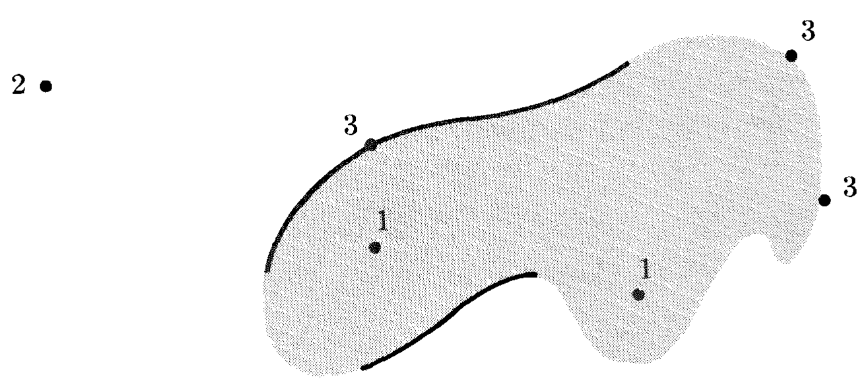

FIGURE 1-2

Those points satisfying (1) constitute the **interior** of A, those satisfying (2) the **exterior** of A, and those satisfying (3) the **boundary** of A. Problems 1-16 to 1-18 show that these terms may sometimes have unexpected meanings.

It is not hard to see that the interior of any set A is open, and the same is true for the exterior of A, which is, in fact, the interior of $\mathbf{R}^n - A$. Thus (Problem 1-14) their union is open, and what remains, the boundary, must be closed.

A collection $\mathcal{O}$ of open sets is an **open cover** of A (or, briefly, **covers** A) if every point $x \in A$ is in some open set in the collection $\mathcal{O}$. For example, if $\mathcal{O}$ is the collection of all open intervals $(a, a + 1)$ for $a \in \mathbf{R}$, then $\mathcal{O}$ is a cover of $\mathbf{R}$. Clearly no finite number of the open sets in $\mathcal{O}$ will cover $\mathbf{R}$ or, for that matter, any unbounded subset of $\mathbf{R}$. A similar situation can also occur for bounded sets. If $\mathcal{O}$ is the collection of all open intervals $(1/n, 1 - 1/n)$ for all integers $n > 1$, then $\mathcal{O}$ is an open cover of $(0,1)$, but again no finite collection of sets in $\mathcal{O}$ will cover $(0,1)$. Although this phenomenon may not appear particularly scandalous, sets for which this state of affairs cannot occur are of such importance that they have received a special designation: a set A is called **compact** if every open cover $\mathcal{O}$ contains a finite subcollection of open sets which also covers A.

A set with only finitely many points is obviously compact and so is the infinite set A which contains 0 and the numbers $1/n$ for all integers n (reason: if $\mathcal{O}$ is a cover, then $0 \in U$ for some open set U in $\mathcal{O}$; there are only finitely many other points of A not in U, each requiring at most one more open set).

Recognizing compact sets is greatly simplified by the following results, of which only the first has any depth (i.e., uses any facts about the real numbers).

1-3 Theorem (Heine-Borel). *The closed interval $[a,b]$ is compact.*

Proof. If $\mathcal{O}$ is an open cover of $[a,b]$, let

$$A = \{x \colon a \le x \le b \text{ and } [a,x] \text{ is covered by some finite number of open sets in } \mathcal{O}\}.$$

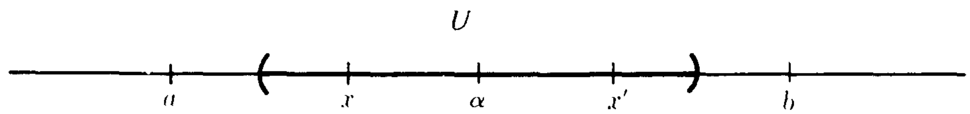

FIGURE 1-3

Note that $a \in A$ and that A is clearly bounded above (by b). We would like to show that $b \in A$. This is done by proving two things about α = least upper bound of A; namely, (1) $\alpha \in A$ and (2) $b = \alpha$.

Since $\mathcal{O}$ is a cover, $\alpha \in U$ for some U in $\mathcal{O}$. Then all points in some interval to the left of α are also in U (see Figure 1-3). Since α is the least upper bound of A, there is an x in this interval such that $x \in A$. Thus $[a,x]$ is covered by some finite number of open sets of $\mathcal{O}$, while $[x,\alpha]$ is covered by the single set U. Hence $[a,\alpha]$ is covered by a finite number of open sets of $\mathcal{O}$, and $\alpha \in A$. This proves (1).

To prove that (2) is true, suppose instead that $\alpha < b$. Then there is a point x' between α and b such that $[\alpha,x'] \subset U$. Since $\alpha \in A$, the interval $[a,\alpha]$ is covered by finitely many open sets of $\mathcal{O}$, while $[\alpha,x']$ is covered by U. Hence $x' \in A$, contradicting the fact that α is an upper bound of A. ∎

If $B \subset \mathbf{R}^m$ is compact and $x \in \mathbf{R}^n$, it is easy to see that $\{x\} \times B \subset \mathbf{R}^{n+m}$ is compact. However, a much stronger assertion can be made.

1-4 Theorem. *If B is compact and $\mathcal{O}$ is an open cover of $\{x\} \times B$, then there is an open set $U \subset \mathbf{R}^n$ containing x such that $U \times B$ is covered by a finite number of sets in $\mathcal{O}$.*

Proof. Since $\{x\} \times B$ is compact, we can assume at the outset that $\mathcal{O}$ is finite, and we need only find the open set U such that $U \times B$ is covered by $\mathcal{O}$.

For each $y \in B$ the point (x,y) is in some open set W in $\mathcal{O}$. Since W is open, we have $(x,y) \in U_y \times V_y \subset W$ for some open rectangle $U_y \times V_y$. The sets V_y cover the compact set B, so a finite number $V_{y_1}, \ldots, V_{y_k}$ also cover B. Let $U = U_{y_1} \cap \cdots \cap U_{y_k}$. Then if $(x',y') \in U \times B$, we have

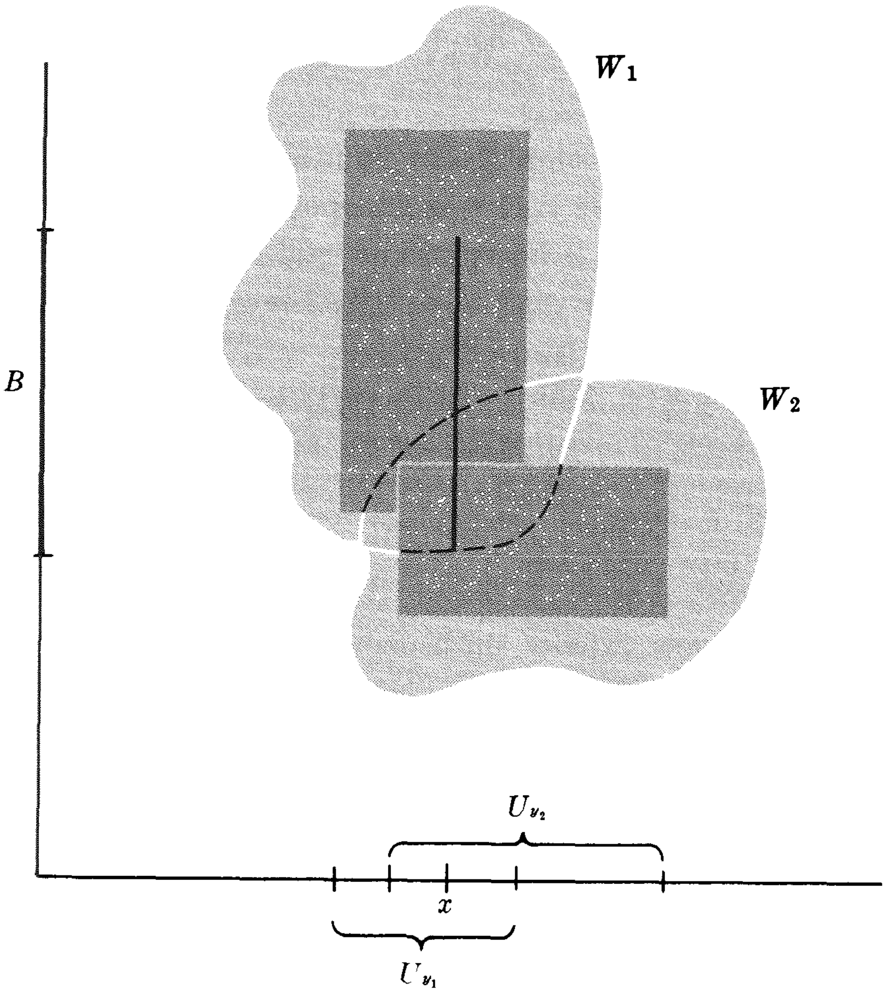

FIGURE 1-4

$y' \in V_{y_i}$ for some i (Figure 1-4), and certainly $x' \in U_{y_i}$. Hence $(x',y') \in U_{y_i} \times V_{y_i}$, which is contained in some W in $\mathcal{O}$. ▌

1-5 Corollary. *If $A \subset \mathbf{R}^n$ and $B \subset \mathbf{R}^m$ are compact, then $A \times B \subset \mathbf{R}^{n+m}$ is compact.*

Proof. If $\mathcal{O}$ is an open cover of $A \times B$, then $\mathcal{O}$ covers $\{x\} \times B$ for each $x \in A$. By Theorem 1-4 there is an open set U_x containing x such that $U_x \times B$ is covered by finitely many sets in $\mathcal{O}$. Since A is compact, a finite number $U_{x_1}, \ldots ,U_{x_n}$ of the U_x cover A. Since finitely many sets in $\mathcal{O}$ cover each $U_{x_i} \times B$, finitely many cover all of $A \times B$. ▌

1-6 Corollary. *$A_1 \times \cdots \times A_k$ is compact if each A_i is. In particular, a closed rectangle in $\mathbf{R}^k$ is compact.*

1-7 Corollary. *A closed bounded subset of* $\mathbf{R}^n$ *is compact.* (The converse is also true (Problem 1-20).)

Proof. If $A \subset \mathbf{R}^n$ is closed and bounded, then $A \subset B$ for some closed rectangle B. If $\mathcal{O}$ is an open cover of A, then $\mathcal{O}$ together with $\mathbf{R}^n - A$ is an open cover of B. Hence a finite number $U_1, \ldots, U_n$ of sets in $\mathcal{O}$, together with $\mathbf{R}^n - A$ perhaps, cover B. Then $U_1, \ldots, U_n$ cover A. ▌

Problems. **1-14.*** Prove that the union of any (even infinite) number of open sets is open. Prove that the intersection of two (and hence of finitely many) open sets is open. Give a counterexample for infinitely many open sets.

1-15. Prove that $\{x \in \mathbf{R}^n : |x - a| < r\}$ is open (see also Problem 1-27).

1-16. Find the interior, exterior, and boundary of the sets

$$\{x \in \mathbf{R}^n : |x| \leq 1\}$$
$$\{x \in \mathbf{R}^n : |x| = 1\}$$
$$\{x \in \mathbf{R}^n : \text{each } x^i \text{ is rational}\}.$$

1-17. Construct a set $A \subset [0,1] \times [0,1]$ such that A contains at most one point on each horizontal and each vertical line but boundary $A = [0,1] \times [0,1]$. *Hint:* It suffices to ensure that A contains points in each quarter of the square $[0,1] \times [0,1]$ and also in each sixteenth, etc.

1-18. If $A \subset [0,1]$ is the union of open intervals (a_i,b_i) such that each rational number in $(0,1)$ is contained in some (a_i,b_i), show that boundary $A = [0,1] - A$.

1-19.* If A is a closed set that contains every rational number $r \in [0,1]$, show that $[0,1] \subset A$.

1-20. Prove the converse of Corollary 1-7: A compact subset of $\mathbf{R}^n$ is closed and bounded (see also Problem 1-28).

1-21.* (a) If A is closed and $x \notin A$, prove that there is a number $d > 0$ such that $|y - x| \geq d$ for all $y \in A$.

(b) If A is closed, B is compact, and $A \cap B = \emptyset$, prove that there is $d > 0$ such that $|y - x| \geq d$ for all $y \in A$ and $x \in B$. *Hint:* For each $b \in B$ find an open set U containing b such that this relation holds for $x \in U \cap B$.

(c) Give a counterexample in $\mathbf{R}^2$ if A and B are closed but neither is compact.

1-22.* If U is open and $C \subset U$ is compact, show that there is a compact set D such that $C \subset$ interior D and $D \subset U$.

FUNCTIONS AND CONTINUITY

A **function** from $\mathbf{R}^n$ to $\mathbf{R}^m$ (sometimes called a (vector-valued) function of n variables) is a rule which associates to each point in $\mathbf{R}^n$ some point in $\mathbf{R}^m$; the point a function f associates to x is denoted $f(x)$. We write $f\colon \mathbf{R}^n \to \mathbf{R}^m$ (read "f takes $\mathbf{R}^n$ into $\mathbf{R}^m$" or "f, taking $\mathbf{R}^n$ into $\mathbf{R}^m$," depending on context) to indicate that $f(x) \in \mathbf{R}^m$ is defined for $x \in \mathbf{R}^n$. The notation $f\colon A \to \mathbf{R}^m$ indicates that $f(x)$ is defined only for x in the set A, which is called the **domain** of f. If $B \subset A$, we define $f(B)$ as the set of all $f(x)$ for $x \in B$, and if $C \subset \mathbf{R}^m$ we define $f^{-1}(C) = \{x \in A : f(x) \in C\}$. The notation $f\colon A \to B$ indicates that $f(A) \subset B$.

A convenient representation of a function $f\colon \mathbf{R}^2 \to \mathbf{R}$ may be obtained by drawing a picture of its graph, the set of all 3-tuples of the form $(x,y,f(x,y))$, which is actually a figure in 3-space (see, e.g., Figures 2-1 and 2-2 of Chapter 2).

If $f,g\colon \mathbf{R}^n \to \mathbf{R}$, the functions $f + g$, $f - g$, $f \cdot g$, and f/g are defined precisely as in the one-variable case. If $f\colon A \to \mathbf{R}^m$ and $g\colon B \to \mathbf{R}^p$, where $B \subset \mathbf{R}^m$, then the **composition** $g \circ f$ is defined by $g \circ f(x) = g(f(x))$; the domain of $g \circ f$ is $A \cap f^{-1}(B)$. If $f\colon A \to \mathbf{R}^m$ is 1-1, that is, if $f(x) \neq f(y)$ when $x \neq y$, we define $f^{-1}\colon f(A) \to \mathbf{R}^n$ by the requirement that $f^{-1}(z)$ is the unique $x \in A$ with $f(x) = z$.

A function $f\colon A \to \mathbf{R}^m$ determines m **component functions** $f^1, \ldots, f^m\colon A \to \mathbf{R}$ by $f(x) = (f^1(x), \ldots, f^m(x))$. If conversely, m functions $g_1, \ldots, g_m\colon A \to \mathbf{R}$ are given, there is a unique function $f\colon A \to \mathbf{R}^m$ such that $f^i = g_i$, namely $f(x) = (g_1(x), \ldots, g_m(x))$. This function f will be denoted $(g_1, \ldots, g_m)$, so that we always have $f = (f^1, \ldots, f^m)$. If $\pi\colon \mathbf{R}^n \to \mathbf{R}^n$ is the identity function, $\pi(x) = x$, then $\pi^i(x) = x^i$; the function π^i is called the ith **projection function.**

The notation $\lim\limits_{x \to a} f(x) = b$ means, as in the one-variable case, that we can get $f(x)$ as close to b as desired, by choosing x sufficiently close to, but not equal to, a. In mathematical terms this means that for every number $\varepsilon > 0$ there is a number $\delta > 0$ such that $|f(x) - b| < \varepsilon$ for all x in the domain of f which satisfy $0 < |x - a| < \delta$. A function $f\colon A \to \mathbf{R}^m$ is called **continuous** at $a \in A$ if $\lim_{x \to a} f(x) = f(a)$, and f is simply called continuous if it is continuous at each $a \in A$. One of the pleasant surprises about the concept of continuity is that it can be defined without using limits. It follows from the next theorem that $f\colon \mathbf{R}^n \to \mathbf{R}^m$ is continuous if and only if $f^{-1}(U)$ is open whenever $U \subset \mathbf{R}^m$ is open; if the domain of f is not all of $\mathbf{R}^n$, a slightly more complicated condition is needed.

1-8 Theorem. *If $A \subset \mathbf{R}^n$, a function $f\colon A \to \mathbf{R}^m$ is continuous if and only if for every open set $U \subset \mathbf{R}^m$ there is some open set $V \subset \mathbf{R}^n$ such that $f^{-1}(U) = V \cap A$.*

Proof. Suppose f is continuous. If $a \in f^{-1}(U)$, then $f(a) \in U$. Since U is open, there is an open rectangle B with $f(a) \in B \subset U$. Since f is continuous at a, we can ensure that $f(x) \in B$, provided we choose x in some sufficiently small rectangle C containing a. Do this for each $a \in f^{-1}(U)$ and let V be the union of all such C. Clearly $f^{-1}(U) = V \cap A$. The converse is similar and is left to the reader. ▌

The following consequence of Theorem 1-8 is of great importance.

1-9 Theorem. ***If $f\colon A \to \mathbf{R}^m$ is continuous, where $A \subset \mathbf{R}^n$, and A is compact, then $f(A) \subset \mathbf{R}^m$ is compact.***

Proof. Let $\mathcal{O}$ be an open cover of $f(A)$. For each open set U in $\mathcal{O}$ there is an open set V_U such that $f^{-1}(U) = V_U \cap A$. The collection of all V_U is an open cover of A. Since A is compact, a finite number $V_{U_1}, \ldots, V_{U_n}$ cover A. Then $U_1, \ldots, U_n$ cover $f(A)$. ▌

If $f\colon A \to \mathbf{R}$ is bounded, the extent to which f fails to be continuous at $a \in A$ can be measured in a precise way. For $\delta > 0$ let

$$\begin{aligned} M(a,f,\delta) &= \sup\{f(x) : x \in A \text{ and } |x - a| < \delta\}, \\ m(a,f,\delta) &= \inf\{f(x) : x \in A \text{ and } |x - a| < \delta\}. \end{aligned}$$

The **oscillation** $o(f,a)$ of f at a is defined by $o(f,a) = \lim_{\delta \to 0}[M(a,f,\delta) - m(a,f,\delta)]$. This limit always exists, since $M(a,f,\delta) - m(a,f,\delta)$ decreases as δ decreases. There are two important facts about $o(f,a)$.

1-10 Theorem. *The bounded function f is continuous at a if and only if $o(f,a) = 0$.*

Proof. Let f be continuous at a. For every number $\varepsilon > 0$ we can choose a number $\delta > 0$ so that $|f(x) - f(a)| < \varepsilon$ for all $x \in A$ with $|x - a| < \delta$; thus $M(a,f,\delta) - m(a,f,\delta) \leq 2\varepsilon$. Since this is true for every ε, we have $o(f,a) = 0$. The converse is similar and is left to the reader. ▮

1-11 Theorem. *Let $A \subset \mathbf{R}^n$ be closed. If $f\colon A \to \mathbf{R}$ is any bounded function, and $\varepsilon > 0$, then $\{x \in A\colon o(f,x) \geq \varepsilon\}$ is closed.*

Proof. Let $B = \{x \in A\colon o(f,x) \geq \varepsilon\}$. We wish to show that $\mathbf{R}^n - B$ is open. If $x \in \mathbf{R}^n - B$, then either $x \notin A$ or else $x \in A$ and $o(f,x) < \varepsilon$. In the first case, since A is closed, there is an open rectangle C containing x such that $C \subset \mathbf{R}^n - A \subset \mathbf{R}^n - B$. In the second case there is a $\delta > 0$ such that $M(x,f,\delta) - m(x,f,\delta) < \varepsilon$. Let C be an open rectangle containing x such that $|x - y| < \delta$ for all $y \in C$. Then if $y \in C$ there is a δ_1 such that $|x - z| < \delta$ for all z satisfying $|z - y| < \delta_1$. Thus $M(y,f,\delta_1) - m(y,f,\delta_1) < \varepsilon$, and consequently $o(y,f) < \varepsilon$. Therefore $C \subset \mathbf{R}^n - B$. ▮

Problems. **1-23.** If $f\colon A \to \mathbf{R}^m$ and $a \in A$, show that $\lim_{x \to a} f(x) = b$ if and only if $\lim_{x \to a} f^i(x) = b^i$ for $i = 1, \ldots, m$.

1-24. Prove that $f\colon A \to \mathbf{R}^m$ is continuous at a if and only if each f^i is.

1-25. Prove that a linear transformation $T\colon \mathbf{R}^n \to \mathbf{R}^m$ is continuous. *Hint:* Use Problem 1-10.

1-26. Let $A = \{(x,y) \in \mathbf{R}^2\colon x > 0 \text{ and } 0 < y < x^2\}$.

(a) Show that every straight line through $(0,0)$ contains an interval around $(0,0)$ which is in $\mathbf{R}^2 - A$.

(b) Define $f\colon \mathbf{R}^2 \to \mathbf{R}$ by $f(x) = 0$ if $x \notin A$ and $f(x) = 1$ if $x \in A$. For $h \in \mathbf{R}^2$ define $g_h\colon \mathbf{R} \to \mathbf{R}$ by $g_h(t) = f(th)$. Show that each g_h is continuous at 0, but f is not continuous at $(0,0)$.

1-27. Prove that $\{x \in \mathbf{R}^n: |x - a| < r\}$ is open by considering the function $f: \mathbf{R}^n \to \mathbf{R}$ with $f(x) = |x - a|$.

1-28. If $A \subset \mathbf{R}^n$ is not closed, show that there is a continuous function $f: A \to \mathbf{R}$ which is unbounded. *Hint:* If $x \in \mathbf{R}^n - A$ but $x \notin$ interior $(\mathbf{R}^n - A)$, let $f(y) = 1/|y - x|$.

1-29. If A is compact, prove that every continuous function $f: A \to \mathbf{R}$ takes on a maximum and a minimum value.

1-30. Let $f: [a,b] \to \mathbf{R}$ be an increasing function. If $x_1, \ldots, x_n \in [a,b]$ are distinct, show that $\sum_{i=1}^{n} o(f,x_i) < f(b) - f(a)$.

2

Differentiation

BASIC DEFINITIONS

Recall that a function $f\colon \mathbf{R} \to \mathbf{R}$ is differentiable at $a \in \mathbf{R}$ if there is a number $f'(a)$ such that

$$(1)\ \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} = f'(a).$$

This equation certainly makes no sense in the general case of a function $f\colon \mathbf{R}^n \to \mathbf{R}^m$, but can be reformulated in a way that does. If $\lambda\colon \mathbf{R} \to \mathbf{R}$ is the linear transformation defined by $\lambda(h) = f'(a) \cdot h$, then equation (1) is equivalent to

$$(2)\ \lim_{h \to 0} \frac{f(a+h) - f(a) - \lambda(h)}{h} = 0.$$

Equation (2) is often interpreted as saying that $\lambda + f(a)$ is a good approximation to f at a (see Problem 2-9). Henceforth we focus our attention on the linear transformation λ and reformulate the definition of differentiability as follows.